Dynamics of Open Quantum Systems III: Redfield Equation and Dissipative Dynamics

Kai Cui

1 Quantum Master Equation

In the previous note, we derived the quantum master equation (QME) for reduced density operator (RDO) dynamics. In interaction picture, the equation is

$$\begin{aligned}\frac{\partial}{\partial t}\hat{\rho}^{(\mathrm{I})}(t) \quad &= \quad -\frac{i}{\hbar}\sum_{\mu}\langle\Phi_{\mu}\rangle_{\mathrm{R}}[\hat{K}_{\mu}^{(\mathrm{I})}(t),\hat{\rho}^{(\mathrm{I})}(t)] \\ &\quad -\sum_{\mu,\nu}\int_{0}^{t}\mathrm{d}\tau\ \Big(C_{\mu\nu}(t-\tau)[\hat{K}_{\mu}^{(\mathrm{I})}(t),\hat{K}_{\nu}^{(\mathrm{I})}(\tau)\hat{\rho}^{(\mathrm{I})}(\tau)] \\ &\qquad - C_{\nu\mu}(\tau-t)[\hat{K}_{\mu}^{(\mathrm{I})}(t),\hat{\rho}^{(\mathrm{I})}(\tau)\hat{K}_{\nu}^{(\mathrm{I})}(\tau)]\Big)\end{aligned} \tag{1}$$

The superscript (I) denotes quantities in the interaction picture. For an arbitrary operator $\hat{A}$ in Schördinger picture,

$$\hat{A}^{(\mathrm{I})}(t)=\hat{U}_{0}^{\dagger}(t)\hat{A}\hat{U}_{0}(t) \tag{2}$$

where $\hat{U}_{0}(t)=e^{-\frac{i}{\hbar}\hat{H}_{0}t}$ is the time evolution operator associated with the unperturbed Hamiltonian $\hat{H}_{0}$. In QME, **we assumed that the system and reservoir is weakly coupled**, and the coupling $\hat{H}_{\mathrm{SR}}$ is treated as a perturbation. Thus $\hat{H}_{0}=\hat{H}_{\mathrm{S}}+\hat{H}_{\mathrm{R}}$, $\hat{U}_{0}(t)=\hat{U}_{\mathrm{S}}(t)\hat{U}_{\mathrm{R}}(t)$. We further assumed that $\hat{H}_{\mathrm{SR}}$ can be factorized to system part $\hat{K}_{\mu}(\boldsymbol{s})$ and reservoir part $\hat{\Phi}_{\mu}(\boldsymbol{Z})$,

$$\hat{H}_{\mathrm{SR}}=\sum_{\mu}\hat{K}_{\mu}(\boldsymbol{s})\hat{\Phi}_{\mu}(\boldsymbol{Z}) \tag{3}$$

with $\boldsymbol{s}$ and $\boldsymbol{Z}$ being coordinates of the system and the reservoir, respectively.

$$C_{\mu\nu}(t)=\frac{1}{\hbar^{2}}\langle\Delta\hat{\Phi}_{\mu}^{(\mathrm{I})}(t)\Delta\hat{\Phi}_{\nu}^{(\mathrm{I})}(0)\rangle_{\mathrm{R}}=\frac{1}{\hbar^{2}}\Big(\langle\hat{\Phi}_{\mu}^{(\mathrm{I})}(t)\hat{\Phi}_{\nu}^{(\mathrm{I})}(0)\rangle_{\mathrm{R}}-\langle\hat{\Phi}_{\mu}\rangle_{\mathrm{R}}\langle\hat{\Phi}_{\nu}\rangle_{\mathrm{R}}\Big) \tag{4}$$

is the reservoir correlation function.

Switching back to Schrödinger picture, QME becomes

$$\boxed{\begin{aligned}\frac{\partial}{\partial t}\hat{\rho}(t) \quad &= \quad -\frac{i}{\hbar}[\hat{H}_{\mathrm{S}}+\sum_{\mu}\langle\Phi_{\mu}\rangle_{\mathrm{R}}\hat{K}_{\mu},\hat{\rho}(t)] \\ &\quad -\sum_{\mu,\nu}\int_{0}^{t}\mathrm{d}\tau\ \Big(C_{\mu\nu}(\tau)[\hat{K}_{\mu},\hat{U}_{\mathrm{S}}(\tau)\hat{K}_{\nu}\hat{\rho}(t-\tau)\hat{U}_{\mathrm{S}}^{\dagger}(\tau)] \\ &\qquad - C_{\nu\mu}(-\tau)[\hat{K}_{\mu},\hat{U}_{\mathrm{S}}(\tau)\hat{\rho}(t-\tau)\hat{K}_{\nu}\hat{U}_{\mathrm{S}}^{\dagger}(\tau)]\Big)\end{aligned}} \tag{5}$$

The first term involving $\langle\Phi_\mu\rangle_{\mathrm{R}}$ leads to reversible dynamics, while the second term involving $C_{\mu\nu}(\tau)$ is responsible for dissipation dynamics. In the second term, the dynamics of $\hat{\rho}(t)$ depends on its dynamic history $\hat{\rho}(t-\tau)$ at all times before t. **The QME is a non-Markovian equation that has memory effects.** Note that the quantum Liouville equation is Markovian. The non-Markovian behavior is a result of only explicitly describing part of the DOFs of the total system.

1.1 Markov Approximation

The characteristic time of the memory effect τ_{mem} is determined by the correlation time of the reservoir correlation functions. If the RDO $\hat{\rho}$ does not substantially change on the time scale of τ_{mem}, then we can assume

$$\hat{\rho}(t-\tau) \approx \hat{\rho}(t) \tag{6}$$

and Eq. (5) becomes Markovian. However, this condition may not always be satisfied. Consider an isolated system S with no coupling to the reservoir. Let ρ_{ab} be its density matrix element in energy representation, then

$$\rho_{ab}(t) = e^{-i\omega_{ab}t}\rho_{ab}(0) \tag{7}$$

where $\omega_{ab} = \frac{E_a - E_b}{\hbar}$. The off diagonal elements are oscillating on the time scale of $1/\omega_{ab}$. This oscillation is also present in the time evolution of a non-isolated system coupled to the reservoir. Thus the condition given in Eq. (6) is only valid when $1/\omega_{ab} \gg \tau_{\mathrm{mem}}$.

For a non-isolated system, we can separate the oscillatory factor $e^{-i\omega_{ab}t}$ from $\rho_{ab}(t)$, and only apply Markov approximation to the remaining part. Let $\tilde{\rho}_{ab}$ be a slowly varying envelope, defined as $\rho_{ab}(t) = e^{-i\omega_{ab}t}\tilde{\rho}_{ab}(t)$, then

$$\rho_{ab}(t-\tau) = e^{-i\omega_{ab}(t-\tau)}\tilde{\rho}_{ab}(t-\tau) \approx e^{-i\omega_{ab}(t-\tau)}\tilde{\rho}_{ab}(t) = e^{i\omega_{ab}\tau}\rho_{ab}(t) \tag{8}$$

Applying similar treatment to the RDO, we have

$$\begin{aligned}
\hat{\rho}(t-\tau) &= \hat{U}_{\mathrm{S}}(t-\tau)\hat{\rho}^{(\mathrm{I})}(t-\tau)\hat{U}_{\mathrm{S}}^{\dagger}(t-\tau) \\
&\approx \hat{U}_{\mathrm{S}}(t-\tau)\hat{\rho}^{(\mathrm{I})}(t)\hat{U}_{\mathrm{S}}^{\dagger}(t-\tau) \\
&= \hat{U}_{\mathrm{S}}^{\dagger}(\tau)\hat{U}_{\mathrm{S}}(t)\hat{\rho}^{(\mathrm{I})}(t)\hat{U}_{\mathrm{S}}^{\dagger}(t)\hat{U}_{\mathrm{S}}(\tau) \\
&= \hat{U}_{\mathrm{S}}^{\dagger}(\tau)\hat{\rho}(t)\hat{U}_{\mathrm{S}}(\tau)
\end{aligned} \tag{9}$$

Substituting this approximation into Eq. (5), and changing the upper limit of the integral from t to ∞,* we have

$$\boxed{\begin{aligned}
\frac{\partial}{\partial t}\hat{\rho}(t) = &-\frac{i}{\hbar}[\hat{H}_{\mathrm{S}} + \sum_\mu \langle\Phi_\mu\rangle_{\mathrm{R}}\hat{K}_\mu, \hat{\rho}(t)] \\
&-\sum_{\mu,\nu}\int_0^\infty \mathrm{d}\tau \left(C_{\mu\nu}(\tau)\left[\hat{K}_\mu, \hat{K}_\nu^{(\mathrm{I})}(-\tau)\hat{\rho}(t)\right] - C_{\nu\mu}(-\tau)\left[\hat{K}_\mu, \hat{\rho}(t)\hat{K}_\nu^{(\mathrm{I})}(-\tau)\right]\right)
\end{aligned}} \tag{10}$$

*The Markov approximation essentially assumes that $\tau_{\mathrm{mem}} \to 0$. At this limit, the integrand is only nonzero when $\tau \to 0$. Thus for every $t > 0$, increasing the upper limit won't change the value of the integral.

where $\hat{K}_\nu^{(\mathrm{I})}(-\tau) = \hat{U}_\mathrm{S}(\tau)\hat{K}_\nu\hat{U}_\mathrm{S}^\dagger(\tau)$. This is the **QME with Markov approximation applied to the slow varying envelope**. To further simplify the expression, we define two new operators:

$$\hat{\Lambda}_\mu = \sum_\nu \int_0^\infty \mathrm{d}\tau\, C_{\mu\nu}(\tau)\hat{K}_\nu^{(\mathrm{I})}(-\tau) \tag{11a}$$

$$\hat{\Lambda}_\mu^{(+)} = \sum_\nu \int_0^\infty \mathrm{d}\tau\, C_{\nu\mu}(-\tau)\hat{K}_\nu^{(\mathrm{I})}(-\tau) \tag{11b}$$

Note that if $\hat{\Phi}_\mu$ and $\hat{K}_\mu$ are Hermitian, then $C_{\mu\nu}^*(\tau) = C_{\nu\mu}(-\tau)$, and $\hat{\Lambda}_\mu^{(+)} = \hat{\Lambda}_\mu^\dagger$. Using these operators, Eq. (10) then becomes

$$\boxed{\frac{\partial}{\partial t}\hat{\rho}(t) = -\frac{i}{\hbar}[\hat{H}_\mathrm{S} + \sum_\mu \langle\Phi_\mu\rangle_\mathrm{R}\hat{K}_\mu, \hat{\rho}(t)] - \sum_\mu \left[\hat{K}_\mu, \hat{\Lambda}_\mu\hat{\rho}(t) - \hat{\rho}(t)\hat{\Lambda}_\mu^{(+)}\right]} \tag{12}$$

Expanding the commutator, we can define the **non-Hermitian effective Hamiltonian for the system**,

$$\hat{H}_\mathrm{S,eff} = \hat{H}_\mathrm{S} + \sum_\mu \hat{K}_\mu \left(\langle\Phi_\mu\rangle_\mathrm{R} - i\hbar\hat{\Lambda}_\mu\right) \tag{13a}$$

$$\hat{H}_\mathrm{S,eff}^{(+)} = \hat{H}_\mathrm{S} + \sum_\mu \left(\langle\Phi_\mu\rangle_\mathrm{R} + i\hbar\hat{\Lambda}_\mu^{(+)}\right)\hat{K}_\mu \tag{13b}$$

Again, when $\hat{\Phi}_\mu$ and $\hat{K}_\mu$ are Hermitian, $\hat{H}_\mathrm{S,eff}^{(+)} = \hat{H}_\mathrm{S,eff}^\dagger$. The Markovian QME expressed in terms of $\hat{H}_\mathrm{S,eff}$ and $\hat{H}_\mathrm{S,eff}^{(+)}$ is

$$\boxed{\frac{\partial}{\partial t}\hat{\rho}(t) = -\frac{i}{\hbar}\left(\hat{H}_\mathrm{S,eff}\hat{\rho}(t) - \hat{\rho}(t)\hat{H}_\mathrm{S,eff}^{(+)}\right) + \sum_\mu \left(\hat{\Lambda}_\mu\hat{\rho}(t)\hat{K}_\mu + \hat{K}_\mu\hat{\rho}(t)\hat{\Lambda}_\mu^{(+)}\right)} \tag{14}$$

The non-Hermitian effective Hamiltonian has complex eigenvalues, and its corresponding time evolution operator is non-unitary. Therefore, in the first term of Eq. (14), acting $\hat{H}_\mathrm{S,eff}$ or $\hat{H}_\mathrm{S,eff}^{(+)}$ from left or right of the RDO changes the norm of the state vectors. However, the second term of Eq. (14) counteracts this, preserveing the normalization of the state vectors and $\hat{\rho}(t)$: $\mathrm{Tr}_\mathrm{S}\hat{\rho}(t) = 1$. This can be proved by taking the trace over the system for Eq. (12). Since the trace of a commutator vanishes, we have

$$\boxed{\frac{\partial}{\partial t}\mathrm{Tr}_\mathrm{S}\hat{\rho}(t) = 0} \tag{15}$$

which indicates the conservation of the total probability.

Now let's consider the time evolution of the total energy of the system S, $E_\mathrm{S} = \mathrm{Tr}_\mathrm{S}\left(\hat{H}_\mathrm{S}\hat{\rho}(t)\right)$. Multiplying $\hat{H}_\mathrm{S}$ to the left of Eq. (12) and taking the trace, we have

$$\frac{\partial E_\mathrm{S}}{\partial t} = -\frac{i}{\hbar}\mathrm{Tr}_\mathrm{S}\left(\hat{H}_\mathrm{S}[\hat{H}_\mathrm{S} + \sum_\mu \langle\Phi_\mu\rangle_\mathrm{R}\hat{K}_\mu, \hat{\rho}(t)]\right) - \sum_\mu \mathrm{Tr}_\mathrm{S}\left(\hat{H}_\mathrm{S}\left[\hat{K}_\mu, \hat{\Lambda}_\mu\hat{\rho}(t) - \hat{\rho}(t)\hat{\Lambda}_\mu^{(+)}\right]\right) \tag{16}$$

This equation can be re-written in terms of $[\hat{H}_{\rm S}, \hat{K}_\mu]$ as

$$\boxed{\frac{\partial E_{\rm S}}{\partial t} = -\frac{i}{\hbar}\sum_\mu \langle\Phi_\mu\rangle_{\rm R}{\rm Tr}_{\rm S}\left([\hat{H}_{\rm S}, \hat{K}_\mu]\hat{\rho}(t)\right) - \sum_\mu {\rm Tr}_{\rm S}\left([\hat{H}_{\rm S}, \hat{K}_\mu](\hat{\Lambda}_\mu\hat{\rho}(t) - \hat{\rho}(t)\hat{\Lambda}_\mu^{(+)})\right)} \tag{17}$$

The first term leads to an oscillation of $E_{\rm S}$, corresponding to the energy reversibly transferring back and forth between the system and the reservoir. The second term causes the dissipation of the system energy. A special case is when $[\hat{H}_{\rm S}, \hat{K}_\mu] = 0$ for all μ, **the dissipation does not change the total energy of the system.** One example of such process is the elastic scattering, which leaves the system energy invariant but changes the phase of the state. Assuming the system has non-degenerate states, then we can express $\frac{\partial E_{\rm S}}{\partial t}$ as

$$\frac{\partial E_{\rm S}}{\partial t} = \sum_a E_a \frac{\partial \rho_{aa}}{\partial t} = 0 \tag{18}$$

Meanwhile, the conservation of total probability requires

$$\sum_a \frac{\partial \rho_{aa}}{\partial t} = 0 \tag{19}$$

These two conditions can only be fulfilled when $\frac{\partial \rho_{aa}}{\partial t} = 0$ for all a, which means **the population of the states does not change**. This process is called **pure dephasing**, as it only destroys the phase relation of the off diagonal elements in the density matrix (coherences), without changing the populations.

2 Redfield Equation

2.1 QME in Energy Representation

Using the eigenstates of $\hat{H}_{\rm S}$ as basis states, we can represent the general, non-Markovian QME in matrix form as

$$\begin{aligned}\frac{\partial}{\partial t}\rho_{ab}(t) \quad = \quad & -i\omega_{ab}\rho_{ab}(t) - \frac{i}{\hbar}\sum_\mu\sum_c \langle\Phi_\mu\rangle_{\rm R}\left(K^\mu_{ac}\rho_{cb}(t) - \rho_{ac}(t)K^\mu_{cb}\right) \\ & - \sum_{\mu,\nu}\sum_{c,d}\int_0^t {\rm d}\tau\, \left\{C_{\mu\nu}(\tau)\left(K^\mu_{ac}K^\nu_{cd}e^{i\omega_{bc}\tau}\rho_{db}(t-\tau) - K^\mu_{db}K^\nu_{ac}e^{i\omega_{da}\tau}\rho_{cd}(t-\tau)\right)\right\} \\ & + \sum_{\mu,\nu}\sum_{c,d}\int_0^t {\rm d}\tau\, \left\{C_{\nu\mu}(-\tau)\left(K^\mu_{ac}K^\nu_{db}e^{i\omega_{bc}\tau}\rho_{cd}(t-\tau) - K^\mu_{db}K^\nu_{cd}e^{i\omega_{da}\tau}\rho_{ac}(t-\tau)\right)\right\}\end{aligned} \tag{20}$$

where $K^\mu_{ab} = \langle a|\hat{K}_\mu|b\rangle$. To simplify the notation, we introduce the **memory matrix**:

$$\boxed{M_{ab,cd}(t) = \sum_{\mu,\nu} C_{\mu\nu}(t)K^\mu_{ab}K^\nu_{cd}} \tag{21}$$

If $\hat{\Phi}_\mu$ and $\hat{K}_\mu$ are Hermitian, we will have

$$M^*_{ab,cd}(t) = \sum_{\mu,\nu} C_{\nu\mu}(-t) K^\nu_{dc} K^\mu_{ba} = M_{dc,ba}(-t) \tag{22}$$

Substituting the general definition of $M_{ab,cd}(t)$ (without invoking the Hermtian assumption), the matrix form of the non-Markovian QME becomes

$$\boxed{\begin{aligned} \frac{\partial}{\partial t}\rho_{ab}(t) \quad = \quad & -i\omega_{ab}\rho_{ab}(t) - \frac{i}{\hbar}\sum_\mu \sum_c \langle \Phi_\mu \rangle_\mathrm{R} \left(K^\mu_{ac}\rho_{cb}(t) - \rho_{ac}(t) K^\mu_{cb} \right) \\ & - \sum_{c,d} \int_0^t \mathrm{d}\tau \left\{ M_{ac,cd}(\tau) e^{i\omega_{bc}\tau} \rho_{db}(t-\tau) - M_{db,ac}(\tau) e^{i\omega_{da}\tau} \rho_{cd}(t-\tau) \right\} \\ & + \sum_{c,d} \int_0^t \mathrm{d}\tau \left\{ M_{db,ac}(-\tau) e^{i\omega_{bc}\tau} \rho_{cd}(t-\tau) - M_{cd,db}(-\tau) e^{i\omega_{da}\tau} \rho_{ac}(t-\tau) \right\} \end{aligned}} \tag{23}$$

By setting $a = b$ and summing over a, and interchange the summation indices a, c, d when necessary, it's easy to show that this equation satisfies the conservation of total probability: $\sum_a \frac{\partial \rho_{aa}}{\partial t} = 0$.

2.2 Asymptotic Solution

When $t \to \infty$, the system should reach its thermal equilibrium with the reservoir, and its density matrix is given by the equilibrium density matrix:

$$\boxed{\lim_{t\to\infty} \rho_{ab}(t) = \delta_{ab} \frac{e^{-E_a/k_\mathrm{B}T}}{\sum_c e^{-E_c/k_\mathrm{B}T}}} \tag{24}$$

We will show that this is the asymptotic solution of Eq. (23). Substituting Eq. (24) into Eq. (23) and setting $t \to \infty$ for the upper limit of the integrals, we need to prove

$$\begin{aligned} 0 \quad = \quad & \frac{i}{\hbar}\sum_\mu \sum_c \langle \Phi_\mu \rangle_\mathrm{R} \left(K^\mu_{ac}\delta_{cb} e^{-E_c/k_\mathrm{B}T} - \delta_{ac} e^{-E_a/k_\mathrm{B}T} K^\mu_{cb} \right) \\ & - \sum_{c,d} \int_0^\infty \mathrm{d}\tau \left\{ M_{ac,cd}(\tau) e^{i\omega_{bc}\tau} \delta_{db} e^{-E_d/k_\mathrm{B}T} - M_{db,ac}(\tau) e^{i\omega_{da}\tau} \delta_{cd} e^{-E_c/k_\mathrm{B}T} \right\} \\ & + \sum_{c,d} \int_0^\infty \mathrm{d}\tau \left\{ M_{db,ac}(-\tau) e^{i\omega_{bc}\tau} \delta_{cd} e^{-E_c/k_\mathrm{B}T} - M_{cd,db}(-\tau) e^{i\omega_{da}\tau} \delta_{ac} e^{-E_a/k_\mathrm{B}T} \right\} \end{aligned} \tag{25}$$

Performing the summation and interchange the indices c, d in some terms, The right side of the equation becomes

$$\begin{aligned} (\ldots) \quad = \quad & \frac{i}{\hbar}\sum_\mu \langle \Phi_\mu \rangle_\mathrm{R} K^\mu_{ab} \left(e^{-E_b/k_\mathrm{B}T} - e^{-E_a/k_\mathrm{B}T} \right) \\ & - \sum_c \int_0^\infty \mathrm{d}\tau \left\{ M_{ac,cb}(\tau) e^{i\omega_{bc}\tau} e^{-E_b/k_\mathrm{B}T} - M_{cb,ac}(\tau) e^{i\omega_{ca}\tau} e^{-E_c/k_\mathrm{B}T} \right\} \\ & + \sum_c \int_0^\infty \mathrm{d}\tau \left\{ M_{cb,ac}(-\tau) e^{i\omega_{bc}\tau} e^{-E_c/k_\mathrm{B}T} - M_{ac,cb}(-\tau) e^{i\omega_{ca}\tau} e^{-E_a/k_\mathrm{B}T} \right\} \end{aligned}$$

$$
\begin{aligned}
= \;& \frac{i}{\hbar}\sum_{\mu}\langle\Phi_\mu\rangle_{\rm R} K^{\mu}_{ab}\left(e^{-E_b/k_{\rm B}T}-e^{-E_a/k_{\rm B}T}\right) \\
&-\sum_c \int_0^{\infty} \mathrm{d}\tau \,\left\{M_{ac,cb}(\tau)e^{i\omega_{bc}\tau}e^{-E_b/k_{\rm B}T}-M_{cb,ac}(\tau)e^{i\omega_{ca}\tau}e^{-E_c/k_{\rm B}T}\right\} \\
&+\sum_c \int_{-\infty}^{0} \mathrm{d}\tau \,\left\{M_{cb,ac}(\tau)e^{i\omega_{cb}\tau}e^{-E_c/k_{\rm B}T}-M_{ac,cb}(\tau)e^{i\omega_{ac}\tau}e^{-E_a/k_{\rm B}T}\right\} \qquad (26)
\end{aligned}
$$

Here we skip the general proof for ρ_{ab} and only focus on the proof for the diagonal elements, where $a=b$. In this case, the equation can be simplified to

$$
\begin{aligned}
(\ldots) \;=\;& -\sum_c \int_0^{\infty} \mathrm{d}\tau \,\left\{M_{ac,ca}(\tau)e^{i\omega_{ac}\tau}e^{-E_a/k_{\rm B}T}-M_{ca,ac}(\tau)e^{i\omega_{ca}\tau}e^{-E_c/k_{\rm B}T}\right\} \\
&+\sum_c \int_{-\infty}^{0} \mathrm{d}\tau \,\left\{M_{ca,ac}(\tau)e^{i\omega_{ca}\tau}e^{-E_c/k_{\rm B}T}-M_{ac,ca}(\tau)e^{i\omega_{ac}\tau}e^{-E_a/k_{\rm B}T}\right\} \\
=\;& \sum_c \int_{-\infty}^{\infty} \mathrm{d}\tau \,\left\{M_{ca,ac}(\tau)e^{i\omega_{ca}\tau}e^{-E_c/k_{\rm B}T}-M_{ac,ca}(\tau)e^{i\omega_{ac}\tau}e^{-E_a/k_{\rm B}T}\right\} \qquad (27)
\end{aligned}
$$

Introducing the Fourier transform of $M_{ab,cd}(t)$,

$$
M_{ab,cd}(\omega)=\int_{-\infty}^{\infty} M_{ab,cd}(t)e^{i\omega t}\,\mathrm{d}t=\sum_{\mu,\nu}C_{\mu\nu}(\omega)K^{\mu}_{ab}K^{\nu}_{cd} \qquad (28)
$$

where $C_{\mu\nu}(\omega)$ is the Fourier transform of the reservoir correlation function, we then only need to prove

$$
\begin{aligned}
0 \;=\;& \sum_c \left\{M_{ca,ac}(\omega_{ca})e^{-E_c/k_{\rm B}T}-M_{ac,ca}(\omega_{ac})e^{-E_a/k_{\rm B}T}\right\} \\
=\;& \sum_c\sum_{\mu,\nu}\left\{C_{\mu\nu}(\omega_{ca})K^{\mu}_{ca}K^{\nu}_{ac}e^{-E_c/k_{\rm B}T}-C_{\mu\nu}(\omega_{ac})K^{\mu}_{ac}K^{\nu}_{ca}e^{-E_a/k_{\rm B}T}\right\} \\
=\;& \sum_c\sum_{\mu,\nu}K^{\mu}_{ca}K^{\nu}_{ac}\left\{C_{\mu\nu}(\omega_{ca})e^{-E_c/k_{\rm B}T}-C_{\nu\mu}(-\omega_{ca})e^{-E_a/k_{\rm B}T}\right\} \qquad (29)
\end{aligned}
$$

The definition of $C_{\mu\nu}(\omega)$ is

$$
C_{\mu\nu}(\omega)=\frac{1}{\hbar^2}\int_{-\infty}^{\infty}\mathrm{Tr}_{\rm R}\left(\hat{R}_{\rm eq}\hat{U}^{\dagger}_{\rm R}(t)\Delta\hat{\Phi}_{\mu}\hat{U}_{\rm R}(t)\Delta\hat{\Phi}_{\nu}\right)e^{i\omega t}\,\mathrm{d}t \qquad (30)
$$

Using the eigenstates of the reservoir Hamiltonian, given by $\hat{H}_{\rm R}|\alpha\rangle=\varepsilon_\alpha|\alpha\rangle$, to calculate the trace, we have

$$
\begin{aligned}
C_{\mu\nu}(\omega) \;=\;& \frac{1}{\hbar^2}\sum_{\alpha,\beta}\int_{-\infty}^{\infty} f(\varepsilon_\alpha)\Delta\Phi^{\mu}_{\alpha\beta}\Delta\Phi^{\nu}_{\beta\alpha}e^{i(\omega+\omega_{\alpha\beta})t}\,\mathrm{d}t \\
=\;& \frac{2\pi}{\hbar^2}\sum_{\alpha,\beta}f(\varepsilon_\alpha)\Delta\Phi^{\mu}_{\alpha\beta}\Delta\Phi^{\nu}_{\beta\alpha}\delta(\omega+\omega_{\alpha\beta}) \qquad (31)
\end{aligned}
$$

and

$$
C_{\nu\mu}(-\omega) \;=\; \frac{2\pi}{\hbar^2}\sum_{\alpha,\beta}f(\varepsilon_\alpha)\Delta\Phi^{\nu}_{\alpha\beta}\Delta\Phi^{\mu}_{\beta\alpha}\delta(\omega-\omega_{\alpha\beta})
$$

$$= \frac{2\pi}{\hbar^2}\sum_{\alpha,\beta} f(\varepsilon_\beta)\Delta\Phi^\mu_{\alpha\beta}\Delta\Phi^\nu_{\beta\alpha}\delta(\omega+\omega_{\alpha\beta}) \tag{32}$$

where $f(\varepsilon_\alpha)$ is the equilibrium Boltzmann distribution reservoir states. Since

$$e^{-\varepsilon_\beta/k_\mathrm{B}T}\delta(\omega+\omega_{\alpha\beta}) = e^{-(\varepsilon_\alpha+\hbar\omega)/k_\mathrm{B}T}\delta(\omega+\omega_{\alpha\beta}) \tag{33}$$

we have

$$\boxed{C_{\mu\nu}(\omega) = \exp\left(\frac{\hbar\omega}{k_\mathrm{B}T}\right)C_{\nu\mu}(-\omega)} \tag{34}$$

This relation obviously proves Eq. (29), as the term in the bracket equals to zero for all c, μ, ν. Thus Eq. (24) is the asymptotic solution for Eq. (23).

2.3 Markov Approximation

Next we apply the Markov approximation to the slow varying envelope of $\rho_{ab}(t)$, by assuming $\rho_{ab}(t-\tau) \approx e^{i\omega_{ab}\tau}\rho_{ab}(t)$ and changing the upper limit of the integrals to ∞, we have

$$\begin{aligned} \frac{\partial}{\partial t}\rho_{ab}(t) \quad = \quad & -i\omega_{ab}\rho_{ab}(t) - \frac{i}{\hbar}\sum_\mu\sum_c \langle\Phi_\mu\rangle_\mathrm{R}\left(K^\mu_{ac}\rho_{cb}(t) - \rho_{ac}(t)K^\mu_{cb}\right) \\ & -\sum_{c,d}\int_0^\infty \mathrm{d}\tau\, \left\{M_{ac,cd}(\tau)e^{i\omega_{dc}\tau}\rho_{db}(t) - M_{db,ac}(\tau)e^{i\omega_{ca}\tau}\rho_{cd}(t)\right\} \\ & +\sum_{c,d}\int_0^\infty \mathrm{d}\tau\, \left\{M_{db,ac}(-\tau)e^{i\omega_{bd}\tau}\rho_{cd}(t) - M_{cd,db}(-\tau)e^{i\omega_{dc}\tau}\rho_{ac}(t)\right\} \end{aligned} \tag{35}$$

The integrals are half Fourier transformation of the memory matrices, which are complex functions. Their real parts describe the irreversible redistribution of populations and coherences, whereas the imaginary parts effectively modify the transition frequency ω_{ab}. Thus, this equation can be formally written as

$$\frac{\partial}{\partial t}\rho_{ab}(t) = -i\tilde{\omega}_{ab}\rho_{ab} + \left(\frac{\partial\rho_{ab}(t)}{\partial t}\right)_\mathrm{diss} \tag{36}$$

where $\tilde{\omega}_{ab}$ is a modified transition frequency due to the mean field coupling between the system and the reservoir and the imaginary parts of the half Fourier transformations. From hereon, we will only focus on the dissipative dynamics described by $\left(\frac{\partial\rho_{ab}(t)}{\partial t}\right)_\mathrm{diss}$, which is determined by the real parts of the half Fourier transformations:

$$\begin{aligned} \left(\frac{\partial\rho_{ab}(t)}{\partial t}\right)_\mathrm{diss} \quad = \quad & -\sum_{c,d}\mathrm{Re}\int_0^\infty \mathrm{d}\tau\, \left\{M_{ac,cd}(\tau)e^{i\omega_{dc}\tau}\rho_{db}(t) - M_{db,ac}(\tau)e^{i\omega_{ca}\tau}\rho_{cd}(t)\right\} \\ & +\sum_{c,d}\mathrm{Re}\int_0^\infty \mathrm{d}\tau\, \left\{M_{db,ac}(-\tau)e^{i\omega_{bd}\tau}\rho_{cd}(t) - M_{cd,db}(-\tau)e^{i\omega_{dc}\tau}\rho_{ac}(t)\right\} \end{aligned} \tag{37}$$

We can define the **damping matrix**,

$$\boxed{\Gamma_{ab,cd}(\omega) = \mathrm{Re}\int_0^\infty \mathrm{d}\tau\, M_{ab,cd}(\tau)e^{i\omega\tau}} \tag{38}$$

Note that

$$\mathrm{Re}\int_0^\infty \mathrm{d}\tau\, M_{ab,cd}(\tau)e^{i\omega\tau} = \mathrm{Re}\int_0^\infty \mathrm{d}\tau\, M^*_{ab,cd}(\tau)e^{-i\omega\tau} \tag{39}$$

If $\hat{\Phi}_\mu$ and $\hat{K}_\mu$ are Hermitian, then Eq. (22) holds. In this case,

$$\boxed{\Gamma_{dc,ba}(-\omega) = \mathrm{Re}\int_0^\infty \mathrm{d}\tau\, M_{ab,cd}(-\tau)e^{i\omega\tau}} \tag{40}$$

Using the definition of the damping matrix and **assuming $\hat{\Phi}_\mu$ and $\hat{K}_\mu$ are Hermitian operators**, we have

$$\begin{aligned}\left(\frac{\partial\rho_{ab}(t)}{\partial t}\right)_{\text{diss}} \quad = \quad -\sum_{c,d}\{&\Gamma_{ac,cd}(\omega_{dc})\rho_{db}(t) + \Gamma_{bd,dc}(-\omega_{dc})\rho_{ac}(t) \\ &- [\Gamma_{db,ac}(\omega_{ca}) + \Gamma_{ca,bd}(-\omega_{bd})]\rho_{cd}(t)\}\end{aligned} \tag{41}$$

We further introduce the **relaxation matrix** or **Redfield tensor**,

$$\boxed{R_{ab,cd} = \delta_{bd}\sum_e \Gamma_{ae,ec}(\omega_{ce}) + \delta_{ac}\sum_e \Gamma_{be,ed}(\omega_{de}) - \Gamma_{db,ac}(\omega_{ca}) - \Gamma_{ca,bd}(\omega_{db})} \tag{42}$$

Then the matrix representation of the Markovian QME has a very compact form:

$$\boxed{\left(\frac{\partial\rho_{ab}(t)}{\partial t}\right)_{\text{diss}} = -\sum_{c,d} R_{ab,cd}\rho_{cd}(t)} \tag{43}$$

This is the **Redfield equation**. Using the definition of the $\hat{\Lambda}$ operator, we can also write $\Gamma_{ab,cd}$ as

$$\Gamma_{ab,cd}(-\omega_{cd}) = \mathrm{Re}\sum_\mu K^\mu_{ab}\Lambda^\mu_{cd} \tag{44}$$

One can show that the same equation can be derived from the matrix representation of Eq. (12).

3 Dissipative Dynamics

Different elements of the Redfield tensor $R_{ab,cd}$ have different effects on the dynamics of the reduced density matrix. In general, the dissipative dynamics of the density matrix can be classified into four catagories, as summarized in Table 1.

Table 1: Classification of the Matrix Elements of $R_{ab,cd}$

Matrix element	Process
$R_{aa,cc}$	population relaxation
$R_{ab,ab}$	coherence dephasing
$R_{ab,cd}$	coherence transfer
$R_{aa,cd}, R_{ab,cc}$	conversion between population and coherence

3.1 Population Relaxation

When $a = b,\ c = d$, the element $R_{aa,cc}$ describes the **evolution of diagonal elements in the reduced density matrix**, which corresponds to population transfer or **population relaxation**. The matrix element is

$$\boxed{R_{aa,cc} = 2\delta_{ac}\sum_e \Gamma_{ae,ea}(\omega_{ae}) - 2\Gamma_{ca,ac}(\omega_{ca}) \equiv \delta_{ac}\sum_e k_{ae} - k_{ca}} \tag{45}$$

where k_{ab} is the transition rate from state $|a\rangle$ to $|b\rangle$, which is defined as

$$\begin{aligned} k_{ab} &= 2\Gamma_{ab,ba}(\omega_{ab}) = 2\mathrm{Re}\int_0^\infty \mathrm{d}\tau\, M_{ab,ba}(\tau)e^{i\omega_{ab}\tau} \\ &= \int_0^\infty \mathrm{d}\tau\, M_{ab,ba}(\tau)e^{i\omega_{ab}\tau} + \int_0^\infty \mathrm{d}\tau\, M^*_{ab,ba}(\tau)e^{-i\omega_{ab}\tau} \\ &= \int_{-\infty}^\infty \mathrm{d}\tau\, M_{ab,ba}(\tau)e^{i\omega_{ab}\tau} \end{aligned} \tag{46}$$

where we have used Eq. (22), which is valid if $\hat{\Phi}_\mu$ and $\hat{K}_\mu$ are Hermitian. Therefore,

$$\boxed{k_{ab} = M_{ab,ba}(\omega_{ab}) = \sum_{\mu,\nu} C_{\mu\nu}(\omega_{ab})K^\mu_{ab}K^\nu_{ba}} \tag{47}$$

The first term in Eq. (45) describes the population transfer from state $|a\rangle$ to all other states, and the second term describes the population transfer from other states to state $|a\rangle$.

Using the explicit expression of $C_{\mu\nu}(\omega)$,

$$C_{\mu\nu}(\omega) = \frac{2\pi}{\hbar^2}\sum_{\alpha,\beta} f(\varepsilon_\alpha)\Delta\Phi^\mu_{\alpha\beta}\Delta\Phi^\nu_{\beta\alpha}\delta(\omega+\omega_{\alpha\beta}) \tag{48}$$

we can write k_{ab} as

$$k_{ab} = \frac{2\pi}{\hbar^2}\sum_{\alpha,\beta}\sum_{\mu,\nu} f(\varepsilon_\alpha)\Delta\Phi^\mu_{\alpha\beta}K^\mu_{ab}\Delta\Phi^\nu_{\beta\alpha}K^\nu_{ba}\delta(\omega_{ab}+\omega_{\alpha\beta}) \tag{49}$$

or

$$k_{ab} = \frac{2\pi}{\hbar}\sum_{\alpha,\beta} f(\varepsilon_\alpha)\left|\langle a\alpha|\Delta\hat{H}_{\mathrm{SR}}|b\beta\rangle\right|^2\delta(E_a+\varepsilon_\alpha-E_b-\varepsilon_\beta) \tag{50}$$

This rate constant has a golden rule-type expression. The transition is caused by the fluctuation of the system-reservoir coupling $\Delta\hat{H}_{\mathrm{SR}} = \hat{H}_{\mathrm{SR}} - \langle\hat{H}_{\mathrm{SR}}\rangle_{\mathrm{R}}$, and the energy is interchanged between the system and the reservoir with the total energy being conserved (resulted from the delta function). Since this transition is accompanied with energy dissipation into the reservoir, it is called the **energy relaxation rate**. Using Eq. (33), one can show that

$$\boxed{k_{ab} = k_{ba}e^{\frac{E_a-E_b}{k_{\mathrm{B}}T}}} \tag{51}$$

This is the **principle of detailed balance**. It ensures that the relaxation will lead to thermal equilibrium.

3.2 Coherence Dephasing

When $a \neq b, a = c, b = d$, the element $R_{ab,ab}$ describes the **damping of the off-diagonal elements in the reduced density matrix**, which corresponds to **coherence dephasing**. The expression of this element is

$$\boxed{R_{ab,ab} = \sum_e \Gamma_{ae,ea}(\omega_{ae}) + \sum_e \Gamma_{be,eb}(\omega_{be}) - \Gamma_{bb,aa}(0) - \Gamma_{aa,bb}(0) \equiv \gamma_{ab}} \tag{52}$$

γ_{ab} is called the **dephasing rate**. It has two contributions, the first two terms corresponds to the population transfer from states $|a\rangle$ and $|b\rangle$ to all other states. Therefore, **energy relaxation is one source of coherence dephasing.** The last two terms is defined by the correlation function at zero frequency. It represents **elastic processes where there is no energy exchange between system and reservoir,** i.e., **pure dephasing**. The pure dephasing rate $\gamma_{ab}^{(\mathrm{pd})}$ is defined as

$$\boxed{\gamma_{ab}^{(\mathrm{pd})} = -\Gamma_{bb,aa}(0) - \Gamma_{aa,bb}(0) = -\sum_{\mu,\nu} C_{\mu\nu}(0) K_{aa}^{\mu} K_{bb}^{\nu}} \tag{53}$$

Substituting Eq. (48), we have

$$\gamma_{ab}^{(\mathrm{pd})} = \frac{2\pi}{\hbar} \sum_{\alpha,\beta} f(\varepsilon_\alpha) \langle a\alpha | \Delta \hat{H}_{\mathrm{SR}} | a\beta \rangle \langle b\beta | \Delta \hat{H}_{\mathrm{SR}} | b\alpha \rangle \delta(\varepsilon_\alpha - \varepsilon_\beta) \tag{54}$$

Clearly, $\gamma_{ab}^{(\mathrm{pd})} = \gamma_{ba}^{(\mathrm{pd})}$.

Using the definitions of $\gamma_{ab}^{(\mathrm{pd})}$ and k_{ab}, we can write

$$\boxed{\gamma_{ab} = \frac{1}{2} \sum_e k_{ae} + \frac{1}{2} \sum_e k_{be} + \gamma_{ab}^{(\mathrm{pd})}} \tag{55}$$

A more commonly used expression in literature is based on dephasing times. T_2 is called the total dephasing time, T_1 is the life time of the states, and T_2^* is the pure dephasing time.

$$T_2 = \frac{1}{\gamma_{ab}}, \quad T_1 = \frac{1}{\sum_e k_{ae} + \sum_e k_{be}}, \quad T_2^* = \frac{1}{\gamma_{ab}^{(\mathrm{pd})}} \tag{56}$$

and

$$\boxed{\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_2^*}} \tag{57}$$

3.3 Secular Approximation

The other matrix elements of the Redfield tensor mix different types of elements of the reduced density matrix, i.e., population and coherence. Under certain conditions, such mixing can be neglected. We first write down the Redfield equation in interaction picture,

$$\left(\frac{\partial \rho_{ab}^{(\mathrm{I})}(t)}{\partial t} \right)_{\mathrm{diss}} = -\sum_{c,d} R_{ab,cd} e^{i(\omega_{ab} - \omega_{cd})t} \rho_{cd}^{(\mathrm{I})}(t) \tag{58}$$

The right hand side of this equation contains multiple contributions which oscillate with frequency $\omega_{ab} - \omega_{cd}$. All contributions where $1/|\omega_{ab} - \omega_{cd}|$ is much smaller than the time step for which the Redfield equation is solved will cancel each other due to destructive interference. We can therefore neglect all these contributions.

Obviously, for population relaxation and coherence dephasing, where $a = b, c = d$ or $a = c, b = d$ respectively, $\omega_{ab} - \omega_{cd} = 0$, thus their contributions can never be neglected. If the system contains degenerate transition frequencies, $\omega_{ab} - \omega_{cd} = 0$ can also hold for other $R_{ab,cd}$ elements not belong to these two categories. An intuitive assumption is that **we only consider the elements of the Redfield tensor for which $\omega_{ab} - \omega_{cd} = 0$.** This is called the **secular approximation** or **rotating wave approximation**. Other contributions are called nonsecular terms.

If we further **assume that $\omega_{ab} - \omega_{cd} = 0$ only holds for population relaxation and coherence dephasing**, we arrived at the **Bloch model**. This is a likely a good approximation for anharmonic systems. With in the Bloch model we can write the Redfield equation as

$$\boxed{\left(\frac{\partial \rho_{ab}(t)}{\partial t}\right)_{\text{diss}} = -\delta_{ab} \sum_c R_{aa,cc}\rho_{cc}(t) - (1 - \delta_{ab}) R_{ab,ab}\rho_{ab}(t)} \tag{59}$$

The dynamics of the diagonal elements and off-diagonal elements of the reduced density matrix are now separated. Their dynamics are determined by the energy relaxation and coherence dephasing rates.